不合拍的风：塔科马大桥的陨落

The Wind That Kept No Time: The Fall of Galloping Gertie



风没按节拍吹，大桥为何还是倒了？

英文原文

On the morning of 7 November 1940, a **sustained** wind of about 40 miles an hour set the Tacoma Narrows Bridge **twisting**. Its **deck tilted** from side to side in ever more **violent oscillations** until most of the main span **tore apart** and **tumbled** into the water. No one died. **Footage** of the **spectacle** would later **fascinate** students worldwide.

The bridge had opened in July 1940 as the third-longest **suspension span** in the world at the time, its roadway **hung from** steel cables. **Girders** only eight feet deep **stiffened** the deck, making it **slender**, **elegant**, and **economical**, but far from **rigid**. Even **faint breezes** made it **heave**, earning the **nickname** Galloping Gertie.

Textbooks have long **invoked resonance**: a **periodic** push that matches an object's natural **frequency** can **amplify** its motion, as when **well-timed shoves propel** a child on a swing. The **analogy** is **intuitive** but **deceptive**. That morning's wind did not **pulse** in time with the bridge; it blew fairly **steadily**.

The **mechanism** was **subtler**. Each twist **altered** the passing airflow, which **reinforced** the next twist: a **feedback** loop that kept **tapping** the wind for energy. In March 1941 a federal **panel** blamed the deck's **flexibility** and wing-like lift. Later analysis called the mechanism flutter; in February 1991 two engineers **asserted** in a physics journal that textbooks had **erred**.

The panel urged **wind-tunnel** tests, and trials on **scaled-down replicas gradually** became **customary** for suspension bridges. A **sturdier replacement** opened in October 1950. Gertie's fall may still **instruct** engineers, if read **accurately**: the wind never kept time with the bridge, the bridge's own motion turned a sustained wind **destructive**.

中文译文

1940年11月7日上午，一股时速约40英里的持续大风使塔科马海峡大桥扭动起来。它的桥面左右倾侧，振荡愈发剧烈，直到主跨大部分断裂、并且跌落到水中。无人丧生。这一奇观的影像后来吸引了世界各地的学生。

这座桥于1940年7月通车，作为当时世界上第三长的悬索桥跨径，其路面悬挂在钢缆上。仅八英尺深的钢梁加固了桥面，使其变得纤细、优雅且经济，但远非刚硬。哪怕轻微的微风也会让它起伏，从而赢得了一个绰号：“奔腾的格蒂”。

教科书长期以来一直援引共振来解释：与物体固有频率相匹配的周期性的推力能放大其运动，就像时机恰到好处的一次次猛推推动秋千上的孩子一样。这个类比直观却具有迷惑性。那天上午的风没有与大桥有节奏地跳动；它吹得相当平稳。

其机理要更为微妙。每一次扭动都改变了掠过的气流，而气流又加强了下一次扭动：这是一个不断从风中汲取能量的反馈回路。1941年3月，一个联邦调查委员会把原因归于桥面的柔性和类似机翼的升力。后来的分析把这一机理称为颤振；1991年2月，两位工程师在一份物理学期刊上断言，教科书出错了。

该委员会力主进行风洞试验，并且对按比例缩小的复制品进行试验，对于悬索桥而言也渐渐成为了惯例。一个更坚固的替代品于1950年10月通车（开放）。格蒂的倒塌或许仍能指导着工程师，如果解读得精准：风从未与大桥合拍，是大桥自身的运动使一股持续的风具有了破坏力。

词汇与表达

sustained adj.
持续
a sustained wind of about 40 miles an

twisting v.
扭动

deck n.
桥面

tilted v.
倾侧

violent adj.
剧烈

oscillations n.
振荡
side in ever more violent oscillations

tore apart phr.
断裂
the main span tore apart and tumbled

tumbled v.
跌落

Footage n.
影像
Footage of the spectacle would later

spectacle n.
奇观

fascinate v.
吸引了

suspension n.
悬索
as the third-longest suspension span

span n.
跨径
as the third-longest suspension span

hung from phr.
悬挂在……上

Girders n.
钢梁

stiffened v.
加固了

slender adj.
纤细

elegant adj.
优雅

economical adj.
经济

rigid adj.
刚硬
and economical, but far from rigid.

词汇与表达（续）

faint adj.
轻微的

breezes n.
微风

heave v.
起伏
Even faint breezes made it heave,

nickname n.
绰号

Textbooks n.
教科书

invoked v.
援引
Textbooks have long invoked resonance:

resonance n.
共振
Textbooks have long invoked resonance:

periodic adj.
周期性的
periodic push that matches an object's

frequency n.
频率
object's natural frequency can amplify

amplify v.
放大
object's natural frequency can amplify

well-timed adj.
时机恰到好处的

shoves n.
猛推

propel v.
推动

analogy n.
类比
The analogy is intuitive but

intuitive adj.
直观
analogy is intuitive but deceptive.

deceptive adj.
具有迷惑性
analogy is intuitive but deceptive.

pulse v.
有节奏地跳动

steadily adv.
平稳

词汇与表达（续）

mechanism n.
机理
The mechanism was subtler.

subtler adj.
更为微妙

altered v.
改变了

reinforced v.
加强了

feedback n.
反馈
a feedback loop that kept tapping the

tapping v.
汲取

panel n.
调查委员会

flexibility n.
柔性
panel blamed the deck's flexibility

asserted v.
断言

erred v.
出错了

wind-tunnel n.
风洞

scaled-down adj.
按比例缩小的
scaled-down replicas gradually became

replicas n.
复制品

gradually adv.
渐渐

customary adj.
惯例
replicas gradually became customary

sturdier adj.
更坚固的

replacement n.
替代品

instruct v.
指导着

accurately adv.
精准

destructive adj.
具有了破坏力
turned a sustained wind destructive.

文章简述

- 事故：1940年，刚通车数月的塔科马大桥在持续大风中剧烈扭转，最终断裂坍塌，无人遇难。
- 隐患：加固桥面的梁较浅，虽然轻巧、省钱，抵抗变形的能力却不足；连微风都能使桥起伏，早有“奔腾的格蒂”之名。
- 纠误：课本常用共振解释事故，好比按秋千的节拍反复推它；但当天的风相当平稳，并未按桥的节拍吹。
- 机制：桥每扭一下就改变周围气流，气流又推动它扭得更厉害，形成越扭越大的循环；这叫颤振，与课本的解释不同。
- 后续：事故推动悬索桥用缩小模型做风洞试验成为惯例，检验桥在风中是否稳定；更坚固的新桥于1950年通车。